The IMDBish **C**onceptual Database **D**esign **D**ocument **(CDD)**

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1 IMDBish

The IMDBish database revolves around *movies*. Movies are produced by various kinds of *people*, which include *writers*, *directors*, *actors*, and *crew members* (members is an umbrella category for those people who participate in production in some other way). Every movie has at least one writer and director, but not every movie has crew members. Actors play *characters* in movies, and can either star or act in a movie. A movie necessitates at least one character, cast member, and star. And to be a person of note in the film industry, that person must have participated in the production of a movie in one of the aforementioned roles.

The IMDBish database records data about *awards* that films have been nominated for and/or won, as well as the award nominations and wins that people receive for their contributions in films. The places where these awards are won and lost are referred to as *events*.

The IMDBish database tracks the *genres* and *sub-genres* of movies. A sub-genre must have a super-genre, but movies don't need to have sub-genres.

It is also relevant where movies can be viewed: *theaters*, and *streamers*.

The opinions of *critics* about movies is also worth tracking, though a critic will not appear in the IMDBish database unless a review of theirs is within the database.

The purpose of this database is to hold information about movies, the people who are involved in their production, the ways in which they can be watched, and any accolades that they have received. The main users of the database are people who are interested in movies and celebrities. No user has elevated privileges that others do not.

This document presents the conceptual schema and the logical relational schema for this database. It is organized in four sections. Section 2 contains the complete description of the conceptual schema following the notation described in the template document. Section 3 lists the most important queries that may be posed to this specific database.

2 Conceptual Schema of the IMDBISH database

The order of presentation of the conceptual schema is:

1. Entities: description and attributes.
2. Relationships: description and attributes (if they have them).
3. The EER diagram.
4. The semantic integrity constraints.

2.1 Entities

The entities defined for this database are:

- ACTOR
- AWARD
- CHARACTER
- CREW_MEMBER
- DIRECTOR
- EVENT
- GENRE
- MOVIE
- PERSON
- STREAMER
- SUB_GENRE
- THEATER
- WRITER

A detailed description of each entity follows.

ACTOR: A person that portrays characters in movies.
Some example instances are: Adrienne Barbeau, Hailee Steinfeld, Michael B. Jordan.
No additional attributes.

AWARD An object signifying meaningful influence in the film landscape.
Some example instances are: Best Performance by an Actress in a Leading Role, Best Motion Picture of the Year, Best Sound.

Attributes:	AwardTitle (text)	(SSPF)
	Category (text)	(SSPF)

CHARACTER: A person or creature in a story, fictional or otherwise.
Some example instances are: Scooby Doo, Castor Troy, Sean Archer

Attributes:	**Name** (text)	(SSPF)
	Description (text)	(SSPF)

CREW MEMBER: A person that contributes to a movie in some way other than acting, directing, writing.
Some example instances are: Ludwig Göransson (composer), Francine Maisler (casting director), Ruth E. Carter (costume designer).
Attribute: PredominantTitle (Derived from occurrences in crew relationship) (SSDF)

DIRECTOR: A person who creates films.
Some example instances are: Ryan Coogler, Claire Denis, Sofia Coppola.
No additional attributes.

EVENT An event where celebrities and movies receive accolades.
Some example instances are: Golden Globes, Grammy Awards, Sundance Film Festival.

Attributes:	**EName** (text)	(SSPF)
	Date (DD/MM/YYYY)	(SSPF)
	Address (Street, City, State, Postal)	(CSPF)

GENRE: A classification of movies, with similar themes, tropes or imagery
Some example instances are: Horror, Action, Mystery.

Attributes:	Name (text)	(SSPF)
	Description (test)	(SSPF)

MOVIE: A moving picture that communicates ideas stories, emotions or feelings.
Some example instances are: Con Air, Face/Off, City of Angels, Snake Eyes.

Attributes:	Title (text)	(SSPF)
	ReleaseDate (DD/MM/YYYY)	(SSPF)
	USCertificate (R/PG/PG13/G/NR)	(SSPF)
	PosterImage (link)	(SMPO)
	Trailer (link)	(SMPO)
	Clip (link)	(SMPO)
	AdditionalPromotionalImages (link)	(SMPO)
	Length (minutes, seconds)	(CSSF)
	Country of Origin	(SSPF)
	Budget (in USD)	(SSPF)
	GrossEarnings (in USD)	(SSPO)
	Revenue (GrossEarnings - Budget)	(SSDO)
	PrimaryLanguage (text)	(SSPF)
	AdditionalLanguages (text)	(SMPO)
	Description (text)	(SSPF)
	FilmingLocations (text)	(SMPF)
	PopularityScore (Derived from user watches)	(SSDO)
	ColorInfo (Color/Black&White/Colorized/ACES)	(SMPO)
	SoundInfo (Text)	(SMPO)

PERSON: A person that makes/made contributions in the film industry. PERSON is a superclass with overlapping, and total participation into its subclasses: ACTOR, DIRECTOR, WRITER, CREW_MEMBER.

Some example instances are: Jacob Elordi, Richard King, Stanley Kubrick.

Attributes:	Name (First, Last)	(CSPF)
	IsCelebrity (true/false)	(SSPF)
	Starmeter (int)	(SSPO)
	PromotionalPhotos (link)	(SMPO)
	Clips (link)	(SMPO)
	BirthDate (DD/MM/YYYY)	(SSPF)
	DeathDate (DD/MM/YYYY)	(SSPO)
	Gender (text)	(SSPO)
	Biography (text)	(SSPO)

REVIEW: A critics review of the film.

Some example instances are: A critic's rating of 1/10 on Seven Samurai with the comment "Loved this".

Attributes:	ReviewTitle (text)	(SSPO)
	PostDate (DD/MM/YYYY)	(SSPF)
	Rating (1-10)	(SSPF)
	Text (text)	(SSPO)
	HelpfulLikes (int)	(SSPO)
	HelpfulDislikes (int)	(SSPO)
	AuthorAndPublication (Author, Publication)	(SCPF)

STREAMER: A product that offers its users the ability to watch and stream content such as TV shows and movies.

Some example instances are: Netflix, Hulu, Disney+.

Attributes:	**Name** (text)	(SSPF)
	SubscriptionPrice (in USD per Month)	(SMPO)
	Website (text)	(SSPF)
	RegionAvailability (text)	(SMPO)

SUB_GENRE: A narrower classification of movies, that is a subdivision of a larger genre

Some example instances are: Vampire Horror, Supernatural Horror, Folk Horror.

Attributes:	Prefix (text)	(SSPF)
	Description (text)	(SSPF)

THEATER: A physical place that routinely shows movies or shows after release.

Some example instances are: Regal Winrock, Cinemark Century Rio Plex 24 and XD.

Attributes:	TheaterName (text)	(SSPF)
	StreetAddress (Street, City, State, Postal)	(CSPF)
	Accessibility (Wheel Chair, Hearing Impaired)	(CSPF)

WRITER: A person who creates screenplays for films.

Some example instances are: Ryan Coogler, Claire Denis, Sofia Coppola.

No additional attributes.

2.2 Relationships

There are 13 regular relationships and 2 identifying relationships in this schema.

nominee Associates a PERSON with an AWARD for which they were nominated but didn't necessarily win.

Attributes:	DateNominated (dd/mm/yyyy)	(SSPF)
	DateWon (same as associated event's date)	(SSPF)
	Won (yes/no)	(SSPF)

wrote Associates a WRITER with a MOVIE for which they wrote the screenplay/script.
No additional attributes.

direct Associates a DIRECTOR with a MOVIE that they directed.
No additional attributes.

crew Associates a CREW_MEMBER with a MOVIE on which they contributed.
No additional attributes.

cast Associates an ACTOR with a MOVIE in which they appear in a role.
No additional attributes.

star Associates an ACTOR with a MOVIE in which they appear as a lead performer.
No additional attributes.

char_in Associates ACTORs with the CHARACTERS that they play in MOVIEs.
No additional attributes.

review_of Associates a REVIEW with the MOVIE which it is a review of. Identifying relationship of the weak entity REVIEW.

genre_of Associates a MOVIE with one or more GENRE classifications that describe its primary tropes and themes.
No additional attributes.

sub_genre_of Associates a MOVIE with its SUB_GENREs.
No additional attributes.

shows Associates a MOVIE with a THEATER where it is being shown.

Attributes:	ShowTimes (hour:minute, (pm/am), dd/mm/yyy)	(CMPF)

available_on Associates a MOVIE with a STREAMER on which it is available.
No additional attributes.

nominated Associates a MOVIE with an AWARD that it has been nominated for.

Attributes:	DateNominated (dd/mm/yyyy)	(SSPF)
	DateWon (same as associated event's date)	(SSDF)
	Won (yes/no)	(CSPF)

given_at Relates an AWARD to the EVENT at which it is presented/awarded.
No additional attributes.

type Associates a GENRE with its known SUB_GENRE(s), if any exist. The identifying relationship of the weak entity SUB_GENRE.
No additional attributes.

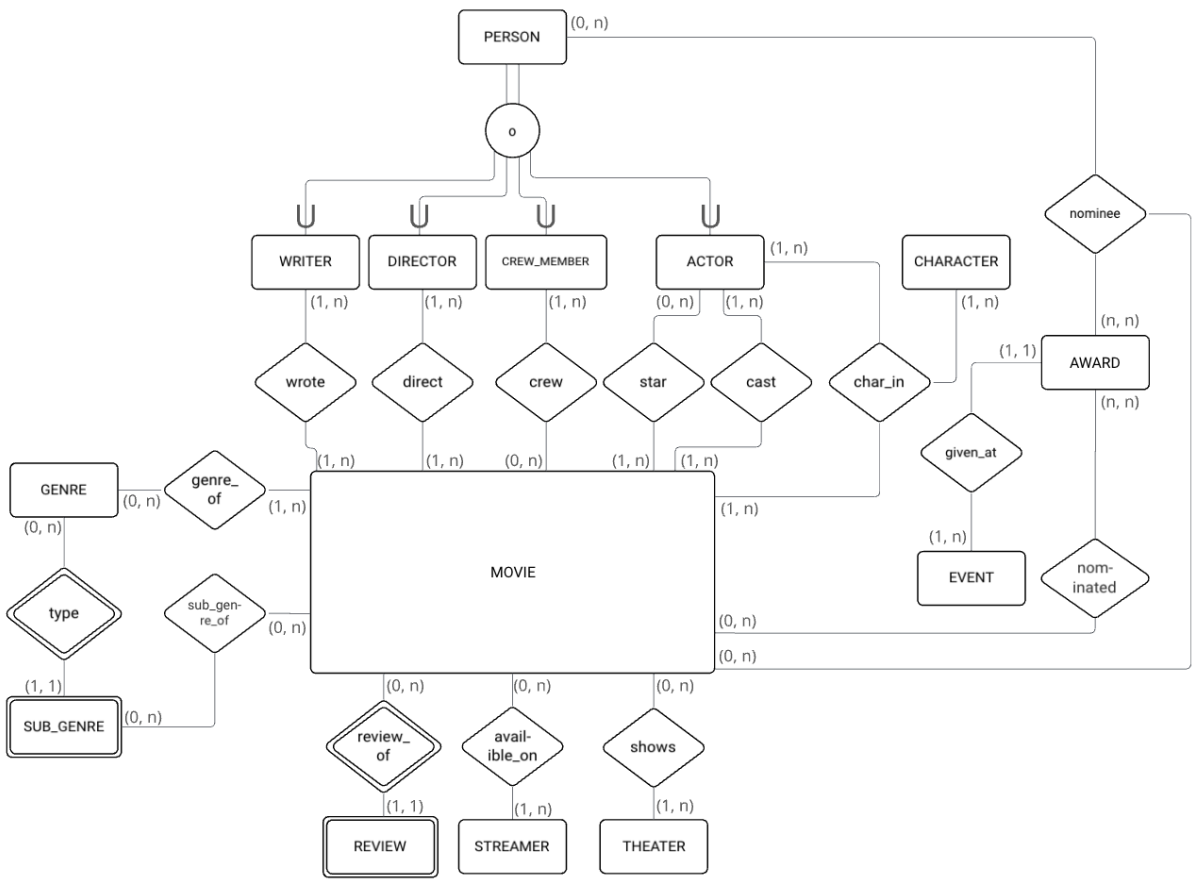

Figure 1: The EER Diagram of IMDB

2.3 Semantic Integrity Constraints

Some semantic integrity constraints associated with this IMDBish database that may not be represented with the EER schema shown are listed in this subsection.

1. A person must participate in a movie to be included in the database, as either a writer, director, actor or crew member.
2. For a movie to have a relationship with a subgenre, the movie must also have a relationship with the sub-genre's parent genre.
3. For all people, `BirthDate` $<$ `DeathDate`.
4. won is a subset of nominated, and similarly recipient is a subset of nominee.
5. star is a subset of cast.
6. In order for a person to be nominated for their part in a movie, they must have worked on that movie.
7. In order for a movie to win or be nominated for an award, its ReleaseDate must be before the DateAwarded and DateNominated respectively.
8. Event Date's and Movie ReleaseDate's cannot be further than 5 years in the future.
9. Review PostDate's cannot be in the future.
10. Review PostDate's cannot be before the ReleaseDate of the movie for which it is a review of.
11. If an event's Date has passed, for all of the awards given at that event, the award must have a winner/recipient.
12. Budget and GrossEarnings of a movie must be greater than zero.
13. AdditionalLanguages of a movie cannot contain the PrimaryLanguage.
14. PopularityScore of a movie is a nonnegative integer.
15. Starmeter of a person is nonnegative.
16. A person must have a BirthDate that is less than all DateAwarded, DateNominated of any awards that person is associated with. Similarly, not able to have DateAwarded and DateNominated more than 10 years after the person's DeathDate.
17. A person must have a BirthDate that is less than the ReleaseDate of any movies which that person participates in. Similarly, a person cannot participate in a movie with a ReleaseDate more than 10 years after the person's DeathDate.
18. The HelpfulLikes and HelpfulDislikes of a review must be nonnegative.

3 Example Queries

A list of the most important queries

1. Provide the list of the most popular movies recently released (within the last month),
2. List the most well received movies by critics within a given time frame.
3. Filter movies based on accessibility, i.e sound and color info for visually or audibly impaired users.
4. For any given person, compile a list of movies that they have participated in, logically sorted according to a user constraint, such as the most popular or, by default, by date.
5. For any given genre, give the list of movies associated with that genre.
6. Find all theaters near a given location that show a specified movie and provide the data for the theater.
7. List all the people who are directors.
8. List the stars of a movie.
9. List all the nominees for all the awards at a particular award show.
10. List the top 10 movies associated with each genre, without repeats.
11. List the top 10 latest trailers that have been released.
12. List all the nominees for upcoming award shows.
13. List all the awards a movie received.
14. List recently added movies on streaming services.
15. List all celebrity's born today.
16. List all movies with a given rating (G, PG, etc).
17. List movies with a release date between some time frame.
18. List top grossing movies with in the last month.

4 Logical Relational Schema (LRS)

PHOTO (*Photo_id, Person_id, Movie_id, File_path, Description, Poster_lag*)
The only attribute that can be Null is Description. Distinct null variables are Person_id and

Movie_id, **one** must be non null.

Domains:	Photo_id	Auto-generated key
	Person_id	ID from the PERSON relation
	Movie_id	ID from the MOVIE relation
	File_path	Path to the photo
	Description	Alternative display/description of image
	Poster_flag	True if image is a poster image, false OW

Foreign Keys: Person_id and Movie_id.

VIDEO (*Video_id File_path, Description, Person_id, Movie_id, Time_minutes*)

The only attribute that can be Null is Description. Distinct null variables are Person_id and Movie_id, **one** must be non null.

Domains:	Video_id	Auto-generated key
	File_path	Path to the video
	Description	Alternative display/description of video
	Person_id	ID from the PERSON relation
	Movie_id	ID from the MOVIE relation
	Time_minutes	length in minutes

Foreign Keys: Person_id and Movie_id.

MOVIE (*Movie_id, Title, Release_date, US_certificate, Length, Country_of_origin, Budget, Gross_earnings, Primary_language, Description, Popularity_score, Color_info, Sound_info*)

The only attributes that can be Null are Budget, Gross_earnings, Popularity_score, Color_info, and Sound_info.

Domains:	Movie_id	Automatically generated integer
	Title	Title of the movie
	Release_date	Movie's release date
	US_certificate	G, PG, PG13, R, NR
	Length	Length of the movie
	Country_of_origin	Movie's Country of Origin
	Budget	How much money movie took to make in USD
	Gross_earnings	How much money movie made in USD
	Primary_language	Language movie was made in
	Description	Movie synopsis
	Popularity_score	How popular the movie is based on user watches
	Color_info	(Color/Black&White/Colorized/ACES)
	Sound_info	Notes about the sound

OTHER_LANGUAGES (*Movie_id, Language*)

No attributes are allowed to be null.

Domains:	Movie_id	ID from MOVIE realtion
	Language	Name of the language

Foreign Keys: Movie_id.

FILMING_LOCATIONS (*Movie_id, Location*)

No attributes are allowed to be null.

Domains:	Movie_id	ID from MOVIE realtion
	Location	Filming location

Foreign Keys: Movie_id.

CREDIT (*Person_id, Movie_id, Actor_flag, Director_flag, Crew_member_flag, Writer_flag, Star_flag, Crew_member_title*)

Distinct null variables are Actor_flag, Director_flag, Crew_member_flag, Writer_flag, Crew_member_title: **one** must be non null and if the Crew_member_flag is non null then the Crew_member_title must also be non null. Star_flag is non-null if and only if Actor_flag is True.

Domains:	Person_id	ID from the PERSON relation
	Movie_id	ID from the MOVIE relation
	Actor_flag	true if the person acted in the movie, false O.W.
	Director_flag	true if the person directed in the movie, false O.W.
	Crew_member_flag	true if the person acted in the movie
	Writer_flag	true if the person wrote the movie, false O.W.
	Star_flag	true if the person is lead role in the movie
	Crew_member_title	if the person is a crew member in this movie, this is their title

Foreign Keys: Person_id, Movie_id.

NOMINEE (*Person_id, Movie_id, Award_id, Date_nominated, Won*)

The only attribute that can be Null is Won.

Domains:	Person_id	ID from the PERSON relation
	Movie_id	ID from the MOVIE relation
	Award_id	ID from the AWARD relation
	Date_nominated	(dd/mm/yyyy)
	Won	True if the award was won, false if lost, null if TBD.

Foreign Keys: Person_id, Movie_id, Award_id.

NOMINATED (*Movie_id, Award_id, Date_nominated, Won*)

The only attribute that can be Null is Won.

Domains:	Movie_id	ID from the MOVIE relation
	Award_id	ID from the AWARD relation
	Date_nominated	(dd/mm/yyyy)
	Won	True if the award was won, false if lost, null if TBD.

Foreign Keys: Movie_id, Award_id.

PERSON (*Person_id, First_name, Last_name, Is_Celebrity, Starmeter, Birthday, Deathday, Gender, Biography*)

The only attributes that can be Null are Deathday, Biography.

Domains:	Person_id	Auto-generated key
	First_name	Person's first name
	Last_name	Person's last name
	IsCelebrity	(true/false)
	Starmeter	Int representing how famous person is
	BirthDate	(DD/MM/YYYY)
	DeathDate	(DD/MM/YYYY)
	Gender	Person's gender
	Biography	Description of person's career

CHARACTER (*Character_id, First_name, Last_name, Description*)

The only attributes that can be Null are Description and Character_last_name.

Domains:	Character_id	Auto-generated key
	Description	Text description of the character
	Character_first_name	First name of character
	Character_last_name	Last name of character

CHARACTER_IN (*Person_id, Movie_id, Character_id*)

No Null attributes.

Domains:	Person_id	ID from the PERSON relation
	Movie_id	ID from the MOVIE relation
	Character_id	ID from the CHARACTER relation

Foreign Keys: Person_id, Movie_id, Character_id.

AWARD (*Award_id, Event_id, Award_title, Category*)

No Null attributes.

Domains:	Award_id	Auto-generated key
	Event_id	ID from the EVENT relation
	Award_title	Title of the award
	Category	Award's category

Foreign Key: Event_id.

REVIEW (*Review_id, Movie_id, Review_title, Post_date, Rating, Text, Helpful_likes, Helpful_dislikes, Author_first_name, Author_last_name, Publication*)

The only attribute that can be Null is Text.

Domains:	Review_id	Auto-generated key
	Movie_id	ID from the MOVIE relation
	Review_title	Synopsis of review
	Post_date	(DD/MM/YYYY)
	Rating	(1-10)
	Text	Body of the review
	Helpful_likes	Likes left by readers of the review
	Helpful_dislikes	Dislikes left by readers of the review
	Author_first_name	First name of writer
	Author_last_name	Last name of writer
	Publication	Endorsement/credibility of author

Foreign Keys: Movie_id.

EVENT (*Event_id, Event_title, Event_date, Event_location*)

No Null attributes.

Domains:	Event_id	Auto-generated key
	Event_title	Name of the event
	Event_date	(DD/MM/YYYY)
	Event_location	Address of the event

GENRE (*Genre_id, Genre_name, Description*)

The only attribute that can be Null is Description.

Domains:	Genre_ id	Auto-generated key
	Genre_name	Title of genre
	Description	The description of the genre

SUB_GENRE (*Sub_genre_id, Genre_id, Sub_genre_prefix, description*)

The only attribute that can be Null is description.

Domains:	Sub_genre_id	Auto-generated key
	Genre_id	ID from the Genre relation
	Sub_genre_prefix	Prefix added to the genre
	Description	Description of the subgenre

Foreign Keys: Genre_id

GENRE_OF (*Movie_id, Genre_id*)

No fields are allowed to be null.

Domains:	Movie_id	ID from the MOVIE relation
	Genre_id	ID from the GENRE relation

Foreign Keys: Genre_id, Movie_id

SUB_GENRE_OF (*Movie_id, Sub_genre_id*)

No fields in this relation may be null.

Domains:	Movie_id	ID from the MOVIE relation
	Sub_genre_id	ID from the SUB_GENRE relation

Foreign Keys: Movie_id, Sub_genre_id.

STREAMER (*Streamer_id, Streamer_name, Website*)

No fields in this relation may be null.

Domains:	Streamer_id	Auto-generated key
	Streamer_name	The name of the streamer
	Website	The streamer's URL

SUBSCRIPTION_PRICE (*Sub_price_id, Streamer_id, Price*)

No fields in this relation may be null.

Domains:	Sub_price_id	Auto-generated key
	Streamer_id	ID from the STREAMER relation
	Price	Cost in USD

Foreign Keys: Streamer_id

REGION_AVAILABILITY (*Region_avail_id, Streamer_id, Region*)

No fields in this relation may be null.

Domains:	Region_avail_id	Auto-generated key
	Streamer_id	ID from the STREAMER relation
	Region	Area where the streamer is available

Foreign Keys: Streamer_id

AVAILABLE_ON (*Movie_id, Streamer_id*)

There are no null attributes.

Domains:		
	Movie_id	ID from the MOVIE relation
	Streamer_id	ID from the STREAMER relation

Foreign Keys: Movie_id, Streamer_id

THEATER (*Theater_id, Theater_name, Street, City, State, Postal, Wheel_chair_accessible, Hearing_impaired_accessible*)

No null attributes.

Domains:		
	Theater_id	Auto-generated key
	Theater_name	Name of Theater
	Street	Name of the street the theater is on
	City	City the theater is in
	State	State the theater is in
	Postal	Postal code of the theater
	Wheel_chair_accessible	True or False
	Hearing_impaired_accessible	True or False

SHOWS (*Movie_id, Theater_id, Showtime*)

There are no null attributes.

Domains:		
	Movie_id	ID from the MOVIE relation
	Theater_id	ID from the THEATER relation
	Showtime	(DD/MM/YYYY/HH/MM/SS)

Foreign Keys: Movie_id, Theater_id

4.1 Additional Integrity Constraints for the relational schema

- An IMAGE's Poster_flag can only be True if Movie_id is non-null.
- A person must be an actor, writer, director or crew member to be included in this database.
- In a NOMINATED relation tuple, if Person_id is non-null and the person is an actor in the movie with the specific Movie_id, then Character_id must also be non-null, and the CHARACTER_IN relation must contain the triple: Person_id, Movie_id, Character_id.
- The Won field in a NOMINATED relation tuple can only be Null if the associated AWARD is given at an EVENT in the future.
- A PERSON can only appear in the CHARACTER_IN relation for a MOVIE, if they appear in the CREDIT relation as an ACTOR.